ROOT CLASSES - MATHEMATICS

Class 10th NCERT Study Material & Formulas

Chapter 10: Circles (Vritt)

Key Concepts & Important Formulas:

* Tangent to a Circle: A line that touches the circle at only one single point.
* Theorem 10.1: Tangent at any point of a circle is perpendicular to radius through point of contact.
* Theorem 10.2: Lengths of tangents drawn from an external point to a circle are equal (PA = PB).
* Number of tangents: No tangent from inside; 1 tangent at point on circle; 2 tangents from outside.

Solved Exam Examples & Practice:

Example: A tangent PQ at point P of circle of radius 5 cm meets line through centre O at Q so OQ = 12 cm. Find PQ.
By Pythagoras: PQ = sqrt(OQ^2 - OP^2) = sqrt(144 - 25) = sqrt(119) cm.